

x-rays. The wavelengths of x-rays are comparable in size to the distance between atoms in most crystals and the repeated pattern of the crystal lattice acts like a diffraction grating for x-rays. In order to disperse x-rays in a spectrometer, a crystal of suitable type can be utilised.

X-RAY ESCAPE PEAK: In a gamma or x-ray spectrum, the peak due to the photoelectric effect in the detector and escape, from the sensitive part of the detector, of the x-ray photon emitted as a result of the photoelectric effect.

X-RAY FLUORESCENCE: The emission of x-rays from excited atoms, produced by the impact of high energy electrons, other particles or primary beam of other X-rays. The technique is used in electron probe microanalysis and as a means of chemical analysis by measuring the wavelengths of the fluorescent x-rays with an x-ray spectrometer.

X-RAY FLUORESCENCE ANALYSIS: A kind of analysis based on the measurement of the energies and intensities of characteristic x-radiation emitted by a test portion during irradiation with electromagnetic radiation.

X-RAY FLUORESCENCE SPECTROMETRY: A technique used to determine the elemental composition of natural and man-made materials such as ceramic, glaze, glass, obsidian, pigments, paint and coins as an aid to determine their source, technology of production and similarity to other examples of comparable style or form. A sample is bombarded with x-rays and the wavelengths of the released energy or fluorescent x-rays are detected and measured.

X-RAY LEVEL: An electronic state occurring as the initial or final state of a process involving the absorption or emission of x-ray radiation. It represents a many-electron state that in the purely atomic case, has total angular momentum ($J = L + S$) as a well-defined quantum number.

X-RAY PHOTOELECTRON SPECTROSCOPY (XPS): Any technique in which the sample is bombarded with x-rays and photoelectrons produced by the sample are detected as a function of energy. **Electron Spectroscopy for Chemical Analysis (ESCA)** refers to the use of this technique to identify elements, their concentrations and their chemical state within the sample.

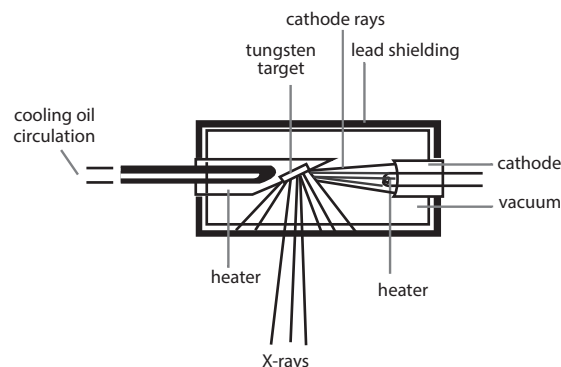
X-RAY SOURCES: Sources of x-radiation from outside the solar system. About a hundred sources have been observed as objects that emit most of their energy in the x-ray region of the electromagnetic spectrum and only a relatively small proportion of their energy in the visible spectrum (within the Galaxy). Most of these x-ray sources seem to be members of a binary system, consisting of one optically visible star and one very compact object. Due to the absorption of x-rays by the Earth's atmosphere, these x-ray sources are mainly visible by x-ray telescopes carried by satellites and space probes.

X-RAY SPECTROSCOPY: The production of X-ray emission and absorption spectra and the use of such spectra for investigating the electronic energy structure of atoms, molecules and solids. X-ray spectroscopy consists of three steps: (i) excitation to produce emission lines characteristic of the elements in the material, (ii) measurement of their intensity and (iii) conversion of X-ray intensity to concentration by a calibration procedure which may include correction for matrix effects.

X-RAY SPECTRUM: A display or graph of the intensity of x-rays, produced when electrons strike a solid object, as a function of wavelengths or some related parameter. It consists of a continuous bremsstrahlung spectrum on which are superimposed groups of sharp lines characteristic of the elements in the target.

X-RAY TUBE: An electronic device for generating x-rays by accelerating electrons to a high energy by an electrostatic field and then suddenly stopping them by striking a metal target either in a

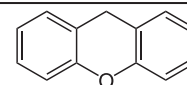
tube containing low pressure gas or in high vacuum. The x-rays are due to the mutual interaction of the fast-moving electrons with the electrons and positively charged nuclei which constitute the atoms of the target.



X-ray tube

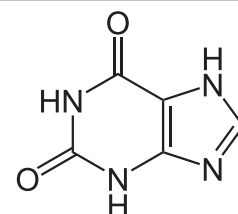
XANTHATES: Organic compounds, usually salts or esters, which contain alternating elements of sulfur-carbon-sulfur [SCS(OR)] group, where R is an organic group. Cellulose xanthate is an intermediate in the manufacture of rayon by the viscose process.

XANTHENE (9H-XANTHENE, 10H-9-OXAANTHRACENE): Yellow crystalline organic compound with the chemical formula $C_{13}H_{10}O$, melting point 101 °C (374 K) and boiling point 310 °C (583 K). It is soluble in diethyl ether. It is used as a fungicide and in making dyes, which includes fluorescein, eosins and rhodamines.



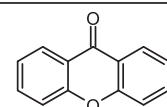
Chemical Structure of Xanthene

XANTHINE (3,7-DIHYDRO-PURINE-2,6-DIONE): Toxic heterobicyclic organic compound with a purine base and the chemical formula $C_5H_4N_4O_2$, found in most human body tissues and fluids and in other organisms. It also occurs in small amounts in potatoes, coffee beans, blood and urine and used industrially as a chemical intermediate. Xanthine is a product on the pathway of purine degradation and created from guanine by guanine deaminase and hypoxanthine by xanthine oxidoreductase. It is subsequently converted to uric acid by the action of the xanthine oxidase enzyme.



Chemical Structure of Xanthine

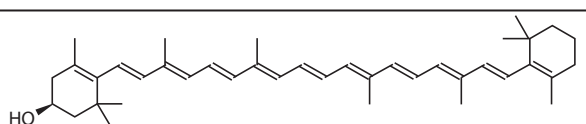
XANTHONE: Plant pigment with the chemical formula $CO(C_6H_4)_2O$, melts at 173 °C (446 K) and sublimates at 350 °C (623 K). It is insoluble in water, soluble in alcohol, chloroform and benzene. It occurs in gentians and other flowers and used as a larvicide (chemical used to kill larvae), as a dye intermediate and in perfumes and pharmaceuticals.



Chemical Structure of Xanthone

XANTHOPHYLLS: A yellow or brown oxygenated carotenoid pigment in plants that occur in all green leaves together with the chlorophyll and responsible for the production of carbohydrates by photosynthesis. Xanthophylls include flavoxanthin, lutein, cryptoxanthin, which is converted into vitamin A, rubixanthin, rhodoxanthin and canthaxanthin. The xanthophylls found in the bodies of animals come from their food, derived from plant sources. For example, the yellow colour of chicken egg yolks, fat and skin comes from ingested xanthophylls in the form of lutein, which is often added to feed for this purpose. The yellow colour of the human macula lutea in the

retina of the eye comes from the lutein and zeaxanthin it contains, derived from the diet. Cryptoxanthin, is an example of a xanthophyll and its chemical structure is given below:



**Chemical Structure of Cryptoxanthin
(an example of Xanthophylls)**

XANTHOPHYLLS CYCLE: A cyclic series of inter-conversions involving certain xanthophylls, which play a vital role in the photo-protection of plant chloroplast.

XANTHOPHYTA (YELLOW-GREEN ALGAE): A phylum of organisms of the kingdom, Protocista that possess carotenoid pigments, which are responsible for their colour, in addition to chlorophylls. Xanthophyte cell walls are produced of cellulose and hemicellulose. Most live in freshwater, but some are found in marine and soil habitats. Xanthophytes occur in a variety of forms: unicellular, colonial, filamentous and siphonaceous; motile cells have two unequal-sized undulipodia. Storage products are oil and polysaccharides.

XENIA: The modification of the form of a fruit or seed by the pollen due to its effect, through double fertilisation, on the nature of the endosperm. For example, in maize the endosperm may show a variety of colours depending on the origin of the pollen.

XENOBIOTIC: Any substance such as pesticide and carcinogen which is foreign to the living systems. It includes drugs such as antibiotics in humans because the human body does not produce them itself, nor are they part of a normal diet. Detoxification of such substance occurs mainly in the liver.

XENOGAMY: Cross-pollination or pollination, which occurs between flowers of different plants.

XENOLITH: A piece of pre-existing rock fragment that occurs as an inclusion within an igneous intrusion. Xenoliths help establish the once molten condition of invading magma capable of incorporating and mixing an assemblage of unrelated rock inclusions.

XENON: A colourless, odourless, highly un-reactive gaseous, non-metallic element with the symbol Xe, melting point -111.9 °C (161 K), boiling point -107°C (166 K), density 5.9 g/L at 0 °C (273 K) and atomic weight 131.29. It belongs to Group 18 (the noble gases) of the Periodic Table. It forms compounds with only fluorine and oxygen. It is found in minute quantities in air and extracted commercially from liquefied air and used in stroboscopic, bactericidal and laser-pumping lamps.

XENOGRAFT (HETEROGRAFT; HETEROLOGOUS GRAFT; HETEROPLASTIC GRAFT): An organ, tissue, skin or cells that is transplanted from one species to another. An example is an organ or tissue that is obtained from a baboon and transplanted into a human body.

XENOPHOBIC ALLIANCE: A union of individual chimpanzees in a group, which challenges intruders who threaten their territory and boundaries.

XENOTRANSPLANTATION: A technique of transplanting animal cells, tissues or organs into humans, for medical purposes.

XENOZOONOSES: An infectious disease transmitted from an animal to a human by transplantation of an animal tissue or organ into human body.

XERIC: Conditions characterised by inadequate supply of water. Xeric conditions exist in arid habitats, extremely cold habitats and salt marshes.

XERO-: A prefix meaning dry.

XEROMORPHIC: A term describing plants or plant parts that are adapted for survival in dry conditions or the structural adaptations

characteristic of xerophytes, such as water-storing stems or leaves. An example is the spiny leaves that reduce surface area and therefore water loss.

XEROPHILOUS: Describing organisms (called xerophiles), which are adapted for growing or living in dry surroundings or conditions with a low availability of water, also known as water activity. Examples of such organisms include cacti, lichens and nematode worms.

XEROPHYTE: A plant such as desert cactus, which is able to grow in very arid conditions because it has adapted to restrict any water loss to the atmosphere. Adaptations include reduction of leaf size, sometimes to spines or scales, a dense covering of hairs over the leaf to trap layers of moist air and shedding leaves at the start of the arid season, water storage cells, sunken stomata, etc.

XEROSERE: The succession of ecological communities which originated in an extremely dry habitat such as a sand desert, sand dunes, a salt desert or a rock desert.

XHTML (EXTENSIBLE HYPertext MARKUP LANGUAGE): A reformulation of HTML and XML based on the HTML 4.0 syntax, but has been modified to follow the guidelines of the Extensible Markup Language (XML). It is designed for network devices and is a method of displaying web pages on network and portable devices.

XIMENIA: A tropical genus of widely distributed tropical shrubs or trees which belongs to the family, Olacaceae and native to the tropics. They have alternate leaves, a persistent calyx, bearded petals and the fruits are drupes. An example is *Ximenia caffra*, also known as the large sourplum tree, is a small deciduous tree or shrub that is thinly branched and can grow up to 6 metres tall. The leathery leaves are simple, alternate and elliptic in shape and occur in clusters on short spur branchlets. The flowers are small, sweet-scented and creamy green and borne in single stem clusters in the axils of the spines or on the branchlets. The green fruit is ellipsoidal with a smooth skin and ripens to an orange or red colour and has a juicy pulp and is edible. The wood of the tree is hard and fine-grained and is used to make handles for tools, utensils or for construction purposes. The seeds are processed into oil. Its various parts such as the roots and bark are used for various medicinal purposes such as the treatment of abscesses, stomach aches, colic, malaria, coughs and bilharzias. *Ximenia americana*, commonly known as yellow plum, false sandalwood or sea lemon, is a semi-scandent shrub or small tree 2 - 7 metres high. The leaves alternate, lanceolate to elliptic in shape, bright green, have an almond smell and contain cyanide. It produces fragrant white, yellow-green or pink flowers that occur in branched inflorescences borne on shortly pedunculate axillary racemes or umbels. The glabrous fruits are globose to ellipsoidal drupes about 3 centimetres long, 2.5 centimetres thick and contains one seed. It contains a juicy pulp when ripe. Its leaves are cooked as a vegetable, but because of its cyanide contents, must be carefully prepared and eaten in small quantities. Like *Ximenia caffra*, its various parts such as the roots and bark are used for various medicinal purposes; the small, hard and dense wood is used as firewood.

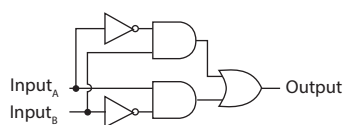
XIPHOID CARTILAGE: The cartilaginous section at the lower end of the sternum, which is not attached to any ribs and gradually ossifies during adult life.

XML (EXTENSIBLE MARKUP LANGUAGE): A specification emanating from the World Wide Web Consortium (W3C) that allows Web designers to create their own language for displaying documents on the Web. XML is an extension to the standard language for creating Web pages, HTML and makes it possible to create websites containing more complex interactivity.

XMODEM: In telecommunications, it is an error detecting FTP (File Transfer Protocol) protocol designed to transmit data in 128-byte blocks via telephone systems at speeds of less than 4,800 bps. If a block is received successfully, a positive (ACK) acknowledgement is returned. If an error is detected, a negative (NAK) acknowledgement

is returned and the block is resent. For faster data transfer rates and error detection the Zmodem is used.

XON/XOFF: In telecommunications, it is a control command used for controlling the flow of data between computers and other devices such as a printer. XON starts or resumes transmission of data and XOFF pauses it. For example, if a computer sends data to a printer faster than the printer can print, the printer sends back an XOFF signal to the computer to stop sending data. When the printer is ready to receive more data it sends an XON signal to the computer to resume sending data. XON and XOFF can be manually activated by control-Q and control-S respectively.



**Exclusive Or (XOR)
Equivalent Circuit And
Logic Gate Truth Table**

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

Logic Gate Truth Table

XOR (EXCLUSIVE OR; EXCLUSIVE DISJUNCTION): A Boolean logic operation for two inputs with the output true only if the two inputs are different; if the two inputs are the same, the output is false. The diagrams show exclusive

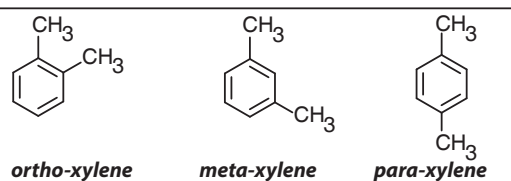
or (XOR) equivalent circuit and logic gate truth table. In computer science, it is a search filter meaning A or B but not both.

XYLAN: A polysaccharide in which the major monosaccharide subunit is xylose. Xylans are abundant components of the hemicelluloses. In monocotyledonous plants the dominant hemicellulose is an arabinoxylan, in which arabinose side chains are attached to a backbone of xylose residues.

XYLEM: The main plant tissue that transports water and mineral nutrients from the roots through the stem and leaves. In flowering plants the major component is the xylem vessel element. It is made up of wide elongated dead cells joined end to end, with the end walls either perforated or totally broken down to allow the rapid transport of water and solutions. Other cells of the xylem tissue are tracheids, fibres and parenchyma.

XYLENE: A colourless, oily, liquid aromatic hydrocarbon with the chemical formula $C_6H_4(CH_3)_2$. It exists in three isomeric forms:

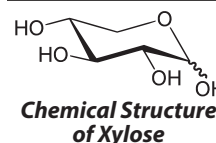
ortho-, *meta*- and *para*- xylene and obtained from coal tar and petroleum. *Ortho*-xylene is 1,2-dimethylbenzene with a melting point of -25°C (248 K) and boils at 144°C (417 K). *Meta*-xylene is 1,3-dimethylbenzene with a melting point of -48°C (91 K) and boils at 139°C (412 K). *Para*-xylene is 1,4-dimethylbenzene with a melting point of 13°C (286 K) and boils at 138°C (411 K). The separation of these three isomers from one another by fractional distillation is difficult because their boiling points are so close together. The *ortho* and *para* isomers are converted to *meta*-xylene by treatment with aluminium trichloride and hydrochloric acid at about 80°C (353 K). The xylenes are often used in the synthesis of other compounds, for example, the xylidenes that are amino derivatives, used in the synthesis of azo dyes and other compounds.



XYLENOL (HYDROXYDIMETHYL BENZENE): Any of six highly toxic isomeric solid aromatic compounds with the chemical formula $C_6H_3(CH_3)_2OH$ and melting point of 20 - 76°C (293-349 K). It is slightly soluble in water and soluble in most organic solvents. An impure mixture of isomers is a liquid mud from coal tar and employed as a solvent. The pure substances resemble phenol in their reactions. It is used as a chemical intermediate, disinfectant, solvent and fungicide and for pharmaceuticals and dyestuffs.

XYLOPHAGOUS: Feeding on or destructive to wood by boring, as in certain molluscs or the larvae of certain insects.

XYLOSE (WOOD SUGAR; D-XYLOSE): A naturally occurring pentose sugar with the chemical formula $C_5H_{10}O_5$, molar mass 150.13 g/mol, density 1.525 g/cm^3 (20°C) and melting point 144 - 145°C . It is found in the form of xylan or as glycosides in the cell walls of many plants such as cherry and maple wood, straw, pecan shell, corn cobs and cotton seed hulls. Since it cannot be digested by humans it does not increase blood sugar levels so it is used in sugar-free foods.



**Chemical Structure
of Xylose**



Y ORGAN: Any of a pair of glands found in the anterior portion of the crustacean body that secretes the hormone ecdysone and takes part in the moulting process.

Y-AXIS: 1. The vertical axis (ordinate) in a two-dimensional coordinate system such as a graph. 2. One of the axes (x , y and z) in the three-dimensional Cartesian coordinate system, conventionally the vertical axis.

Y-CHROMOSOME: The smaller of the two sex chromosomes (the larger being the X-chromosome), which is associated with male characteristics in mammals, occurring paired in the male with the other type of sex chromosome (X), which carries far more genes.

Y-COORDINATE: The position of a point along the y -axis of a two- or three-dimensional coordinate system, which is equal to the directed distance of a point from the x -axis in a two dimensional system or from the plane of the x - and z - axes in a three-dimensional coordinate system. It is measured along a line parallel to the y -axis.

Y-INTERCEPT: The point at which a line crosses the y -axis.

Y2K BUG (YEAR 2000 BUG): An infamous bug that was thought would affect software at the change of the millennium. It is a defect in the code of a computer program caused when a year is represented by its last two digits only and the program interprets that year as falling inclusively between 1900 and 1999 instead of between 2000 and 2099.

YABA VIRUS: A family of deoxyribonucleic acid-containing animal viruses that is characterised by its ability to replicate in the cell cytoplasm. It is a poxvirus of African monkeys, which can infect all primates. The genus contains three members, namely tanapox virus, yaba-like disease virus and yaba monkey tumour virus.

YAGI AERIAL (YAGI-UDA ANTENNA; YAGI ANTENNA; YAGI): A directional antenna array commonly utilised for television and radio telescopes. It is made up of a dipole or two, parallel reflector and a series of closely spaced directors of 0.15 - 0.25 wavelength apart in front of the dipole. When it is used for reception this arrangement focuses the incoming signal on the dipole. For transmission, the output of the dipole is strengthened by the directors.

YAHOO! (YET ANOTHER HIERARCHICAL OFFICIOUS ORACLE!): A popular Search Engine and one of the Internet's leading search engines. It was created in 1994 by two Stanford University postgraduate electrical engineering students, David Filo and Jerry Yang. It is also the largest Web portal, providing links to thousands of other websites. These links include sites from the Yahoo! Directory as well as news stories that are updated several times a day. Other services provided include Yahoo! Finance, Yahoo! Shopping, Yahoo! Games, Yahoo! Mail, Yahoo! Messenger, etc.

YAM: A group of tropical climbing plants belonging to the genus *Dioscorea* of the family Dioscoreaceae or their large edible starchy tubers. Members of the genus are monocots and are native to tropical regions throughout the world. Yam tubers, can grow up to 2.5 metres long and weigh up to 70 kilogrammes. *Dioscorea rotundata*, the "white yam" and *Dioscorea cayenensis*, the "yellow yam", are native to Africa

and are the most important cultivated species. The water yam is classified as *Dioscorea alata*.

YANG-MILLS THEORY: A theory of nuclear forces between nucleons based on the hypothesis that they can be derived by imposing local isospin invariance. This implies that the interaction must occur through the exchange of three massless vector bosons. The theory was formulated in 1953 by Wolfgang Pauli, the first consistent generalisation of the five-dimensional theory of Kaluza, Klein, Fock and others to a higher dimensional internal space.

YARD: A fundamental unit of length in both the U.S. Customary System and the British Imperial System, which is equal to 3 feet or 36 inches. In the SI system, it is equal to 0.9144 metre.

YARROW (MILFOIL): A common wild plant, *Achillea millefolium*, with finely divided leaves. It is used in salads or chopped to replace chervil or parsley as a garnish. It can cause a disagreeable taste or taints in milk, when grazed by cows.

YATES CORRECTION: A continuity correction applied to a chi-squared contingency table test, which involves diminishing the numerical size of the difference between observed and expected values by 0.5 before squaring the chi-square statistic. It is calculated according to Frank

Yates, an English statistician as $\chi^2_{Yates} = \sum_{i=1}^N \frac{(|O_i - E_i| - 0.5)^2}{E_i}$, where O_i

is an observed frequency E_i is an expected (theoretical) frequency, asserted by the null hypothesis and N is the number of distinct events.

YAWS (FRAMBESIA): An infectious tropical chronic disease caused by the bacterium, *Treponema pertenue* and marked by skin eruptions and later by joint pains. It mainly affects children. The disease is widely distributed throughout the tropical regions of the world. Infection is transmitted by direct contact with the discharge from lesions of yaws victims. Yaws may cause severe disabling deformities and disfigurement.

YEAR: 1. The period of time that it takes a planet such as Earth or Mars to make a complete revolution around the Sun. Earth takes 365 days, 5 hours, 49 minutes and 12 seconds of mean solar time to complete one revolution around the Sun. In the Gregorian calendar, the year begins on January 1 and ends on December 31 and is divided into 12 months, 52 weeks and 365 or 366 days. The **civil year** or **calendar year**, is the measure of time in the Gregorian calendar, on which the calendar is based which consists of an average of 365.25 mean solar days or a period of 365 days (or 366 in a leap year), measured from January 1 to December 31. The Martian year takes 687 of Earth days. 2. **Astronomical year**, also called, **solar year**, **equinoctial year** or **tropical year**, is the time between one vernal equinox and the next or successive spring or autumnal equinoxes or winter or summer solstices. It is roughly 365 days, 5 hours, 48 minutes and 46 seconds in length. This period thus marks the regular cycle of the seasons. 3. A **sidereal year** is the time it takes for the Sun to return to the same position with respect to the stars. It is 23 hours 56 minutes and 4.09 seconds long. Due to the precession of the equinoxes the sidereal year is about 20 minutes longer than the tropical year. 4. A **lunar calendar** is based on cycles of the lunar phases. Because there are slightly more than twelve lunations (synodic months) in a solar year, the period of 12 lunar months (354.37 days) is sometimes referred to as a **lunar year**.

YEARLING: An animal aged between one and two years.

YEAST: A group of microscopic, one-celled fungi belonging to the class Hemiascomycetae and to the phylum Ascomycota, which produces asexually by budding and sexually by fusion of cells. The cells making up the thallus occur in pairs, in groups of three or in straight or branched chains consisting of as many as 12 or more cells. Yeasts are widespread in nature, occurring in the soil and on